Dual-Dynamic Cross-Modal Interaction Network for Multimodal Remote Sensing Object Detection

Wei Bao, Meiyu Huang, Jingjing Hu, and Xueshuang Xiang

***Abstract*— Multimodal remote sensing object detection (MM-RSOD) holds great promise for around-the-clock applications. However, it faces challenges in effectively extracting complementary features due to the modality inconsistency and redundancy. Inconsistency can lead to semantic-spatial misalignment, while redundancy introduces uncertainty that is specific to each modality. To overcome these challenges and enhance complementarity exploration and exploitation, this article proposes a dual-dynamic cross-modal interaction network (DDCINet), a novel framework comprising two key modules: a dual-dynamic cross-modal interaction (DDCI) module and a dynamic feature fusion (DFF) module. The DDCI module simultaneously addresses both modality inconsistency and redundancy by employing a collaborative design of channel-gated spatial cross-attention (CSCA) and cross-modal dynamic filters (CMDFs) on evenly segmented multimodal features. The CSCA component enhances the semantic-spatial correlation between modalities by identifying the most relevant channel–spatial features through cross-attention, addressing modality inconsistency. In parallel, the CMDF component achieves cross-modal context interaction through static convolution and further generates dynamic spatial-variant kernels to filter out irrelevant information between modalities, addressing modality redundancy. Following the improved feature extraction, the DFF module dynamically adjusts interchannel dependencies guided by modal-specific global context to fuse features, achieving better complementarity exploitation. Extensive experiments conducted on three MM-RSOD datasets confirm the superiority and generalizability of the DDCINet framework. Notably, our DDCINet, based on the RoI Transformer benchmark and ResNet50 backbone, achieves 78.4% mAP50 on the DroneVehicle test set and outperforms state-of-the-art (SOTA) methods by large margins.**

***Index Terms*— Channel-gated spatial cross-attention (CSCA), cross-modal dynamic filters (CMDFs), dual-dynamic cross-modal interaction (DDCI), multimodal remote sensing object detection (MM-RSOD).**

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I. Introduction

REMOTE sensing object detection (RSOD) aims to identify and precisely locate object instances in images, playing a pivotal role in a broad range of applications, including intelligent monitoring, urban planning, port management, and military reconnaissance [1], [2], [3]. Substantial progress has been achieved in single-modal RSOD based on optical imagery owing to the wealth of edge details, textures, and color information. Nevertheless, single-modal RSOD is limited by insufficient data, particularly under challenging conditions such as poor illumination (nighttime) and various adverse weather conditions (rain, dust, and fog). Recently, researchers have attempted to leverage diverse sensors to assist RSOD because different modalities can provide rich complementary information through optical images [4]. For instance, infrared cameras can generate distinct silhouettes of objects even in low-light conditions, thanks to their capability to capture radiated heat [5]. Synthetic aperture radar (SAR) is an active microwave remote sensing imaging radar with the ability to operate in all-day and all-weather conditions, displaying unique scattering characteristics for various objects [6], [7]. Multimodal RSOD (MM-RSOD) has attracted significant attention, and how to explore and leverage complementary information [8], [9] in MM-RSOD to obtain more accurate results remains a challenging issue.

With the rapid development of deep learning, MM-RSOD has gradually evolved into a data-driven paradigm. This paradigm employs convolutional neural networks (CNNs) [10] or Transformers [11] to extract and fuse multimodal features by leveraging complementary information. Methods such as Halfway Fusion [12] and YOLOrs [13] demonstrate that aggregating middle-level and multiscale features is particularly effective for multimodal object detection. However, these methods suffer from inadequate extraction and fusion of complementary features due to two significant challenges in multimodal remote sensing images: modality inconsistency and redundancy, as depicted in Fig. 1. On the one hand, multimodal images often exhibit modality inconsistency [14] due to significant differences in imaging mechanisms, resulting in semantic and spatial misalignment. This misalignment manifests as misclassification and mislocalization of the same object across different modalities, which complicates the process of feature fusion. On the other hand, modality redundancy [15] refers to the uncertainty present in multimodal data. Optical images tend to be uncertain under poor

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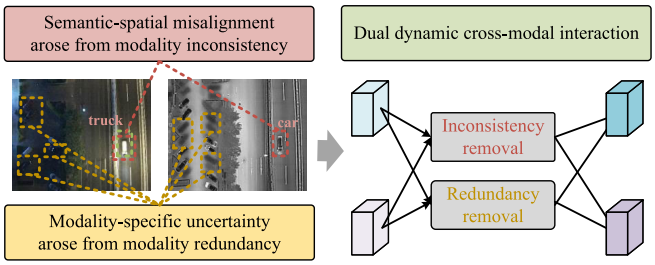


Fig. 1. MM-RSOD faces two major challenges: semantic-spatial misalignment arising from modality inconsistency and modality-specific uncertainty resulting from modality redundancy. The proposed DDCI module effectively addresses these challenges dynamically and simultaneously.

illumination, while infrared images exhibit ghost shadows that appear similar to targets. Modality uncertainty inevitably introduces unnecessary information into the feature fusion process. In summary, modality inconsistency and redundancy significantly impede the extraction of complementary features, which results in suboptimal feature aggregation and inferior detection performance.

Recent advancements have focused on mitigating either modality inconsistency or redundancy to extract complementary features effectively. Regarding modality inconsistency, AR-CNN [16], TSFADet [17], and CAGT-net [18] design spatial alignment detection heads relying heavily on weakly aligned annotations and lacking flexibility. ICAFusion [19] and $\mathbf{C}^2$Former [20] address spatial miscalibration by learning cross-modality relationships. CALNet [21] analyzes contextual information from multimodal analogous pixels to tackle semantic misalignment. DATFF [22] further focuses on both semantic and spatial misalignment via a dual-cross-transformer architecture. These methods mitigate modal heterogeneity through cross-modal global attention but they lack effective aggregation of local information when adopting coarse spatial–channel reduction to limit computational complexity. As for modality redundancy, BAAnet [15] and CMFR [23] leverage cross-modal spatial-wise and channel-wise rectification to filter out modality-specific noisy information. MIN [24] mitigates the effects of clouds in optical images via simple feature fusion, while MoCG [25] addresses speckle noise and foreshortening in SAR images by utilizing global complementary information from optical features. These methods pay more attention to global and static noise but tend to lack adaptability in more complex scenarios where local and dynamic contextual noisy information is prevalent. Different from the above approaches that enhance the capability of extracting complementary features from a singular perspective, CLGNet [26] further emphasizes rectifying spatial misalignment and reducing redundant information. However, it lacks the dynamic and collaborative use of multimodal inputs, which are urgently needed for cross-modal learning. In summary, most existing MM-RSOD methods improve complementary feature extraction from a single perspective, highlighting a gap in integrating a unified, dynamic, and collaborative framework to simultaneously address modality inconsistency and redundancy.

Based on the preceding discussions, we propose a novel framework, a dual-dynamic cross-modal interaction network (DDCINet), to enhance the extraction and fusion of complementary features for MM-RSOD. As shown in Fig. 2, the DDCINet consists of a dual-dynamic cross-modal interaction (DDCI) module and a dynamic feature fusion (DFF) module. The former can dynamically mitigate modality inconsistency and redundancy for better complementarity exploration, while the latter can dynamically fuse multimodal features for better complementarity exploitation. More specifically, the DDCI module performs dual-cross-modal interaction to mitigate modality inconsistency and redundancy (as depicted in Fig. 1) by applying the channel-gated spatial cross-attention (CSCA) block and the cross-modal dynamic filters (CMDFs) block on evenly segmented multimodal features. Initially, two modality input features are split into two halves segments, respectively. Subsequently, one-half of the multimodal features are processed by the CSCA block, while the other half is processed by the CMDF block. Both blocks leverage information from the other modality to enhance complementary feature extraction for the current modality. The CSCA component enhances the semantic-spatial correlation between modalities by identifying the most relevant channel–spatial features through cross-attention, addressing modality inconsistency. In parallel, the CMDF component achieves cross-modal context interaction through static convolution and further generates dynamic spatial-variant kernels to filter out irrelevant information between modalities, addressing modality redundancy. The resulting outputs from each modality are dynamically fused together. This straightforward and efficient "split-interact-fuse" design empowers the network with comprehensive complementarity exploration and achieves better complementary features. Following the improved feature extraction, the DFF module dynamically adjusts interchannel dependencies guided by modal-specific global context to fuse features, achieving better complementarity exploitation. We evaluate the DDCINet on three representative MM-RSOD datasets: DroneVehicle [27], VEDAI [28], and OGSOD [29]. Our extensive experimental results demonstrate the superiority and generalizability of our method, which surpasses existing state-of-the-art (SOTA) methods by large margins. In summary, the main contributions of our work can be summarized as follows.

1) We summarize two challenges that hinder the extraction and fusion of complementary features for MM-RSOD: modality inconsistency usually leading to semantic-spatial misalignment and modality redundancy inevitably introducing modality-specific uncertainty.
2) We propose a novel framework, the DDCINet which includes the DDCI module and DFF module, for better extraction and fusion of complementary features. The effectiveness has also been thoroughly confirmed through extensive experiments.
3) The DDCI module is proposed to simultaneously address modality inconsistency and redundancy by collaboratively applying the CSCA and CMDF on evenly segmented multimodal features, thereby improving complementarity exploration.

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4) The DFF module is proposed to fuse features via dynamically adjusting interchannel dependencies guided by modal-specific global context, achieving better complementarity exploitation.

The rest of this article is organized as follows. Section II introduces some related works, and Section III introduces our methods in detail. Section IV provides the experimental settings and results analysis. Finally, some conclusions and future works are drawn in Section V.

II. Related Work

A. Remote Sensing Object Detection

RSOD aims to identify and precisely locate each object instance in images. Most existing methods adapt common object detection frameworks developed for natural images—such as Faster R-CNN [30], RetinaNet [31], YOLO [32], and DETR [33]—to handle arbitrarily oriented objects by incorporating additional orientation regression. Furthermore, advanced detectors like $\mathbf{S}^2$A-Net [34], RoI Transformer [35], and Oriented R-CNN [36] have been designed to enhance oriented bounding box representation. Despite these significant advancements, single-modal detectors often suffer from limited information and are poorly suited for processing multimodal remote sensing images.

B. Multimodal Object Detection

Traditional multimodal object detectors [12], [13] typically integrate multimodal features through simple addition or concatenation. For example, Halfway Fusion [12] demonstrates that aggregating middle-level features is particularly effective, while YOLOrs [13] fuses multiscale features to enhance vehicle detection across various scales. More advanced approaches focus on addressing modality inconsistency or redundancy to improve the extraction of complementary features. In terms of inconsistency removal, AR-CNN [16] introduces the region feature alignment (RFA) module to tackle spatial misalignment. TSFADet [17] and CAGT-net [18] extend RFA to multimodal remote sensing vehicle detection. However, RFA heavily relies on the computationally expensive ROI-Align operation and weakly aligned annotations, which limits its applicability. ICAFusion [19] and $\mathbf{C}^2$Former [20] address spatial miscalibration by learning cross-attention relationships between modalities. CALNet [21] analyzes contextual information from multimodal analogous pixels to tackle semantic misalignment. DATFF [22] offers a more comprehensive approach by focusing on both semantic and spatial misalignment through a dual-cross-transformer architecture. However, these methods often struggle with the effective aggregation of local information due to their reliance on coarse spatial–channel reductions to manage computational complexity. On the redundancy removal front, BAAnet [15] and CMFR [23] employ cross-modal spatial–channel attention to filter out modality-specific noisy information. MIN [24] uses a simple fusion strategy to mitigate the effects of clouds in optical images. MoCG [25] addresses speckle noise and foreshortening in SAR images by utilizing global complementary information from optical features. However, these methods often lack adaptability in more complex scenarios characterized by diverse and dynamic contextual noise. CLGNet [26], which is most similar to our work, aims to rectify spatial misalignment and reduce redundant information. However, it lacks the dynamism and collaborative approach necessary for effective integration of multimodal information in cross-modal learning. In contrast, our proposed method presents a unified and collaborative framework designed to dynamically and simultaneously address both modality inconsistency and redundancy, leading to enhanced extraction and aggregation of complementary features.

C. Attention Mechanism

The attention mechanism has demonstrated its ability to enhance the representational capacity of models across various tasks in Computer Vision and Natural Language Processing. For instance, SEnet [37] adapts channel-wise feature responses adaptively, while CBAM [38] sequentially extracts cross-channel and spatial information. GCNet [39] aggregates features from all positions to form a global context feature, subsequently modeling channel-wise interdependencies. Transformers [11] capture long-range global context by computing each query position's response as a weighted aggregation of features across all positions, irrespective of their distance. The ViT [40] splits images into tokens through patch embedding operations to facilitate token-wise dependency modeling for computer vision tasks. PVT [41], Swin Transformer [42], and TransXnet [43] focus on reducing the computational cost associated with processing high-resolution inputs. Additionally, the cross-attention mechanism in Transformers is particularly well-suited for cross-modal interactions in multimodal interpretation tasks. The proposed CSCA module is inspired by PVT [41] and SEnet [37], enabling the integration of global spatial and channel information from another modality more flexibly and efficiently.

D. Dynamic Filters

Dynamic filters [44] significantly enhance the effectiveness of attention mechanisms by adaptively extracting features based on input variations. For example, CondConv [45] and DyConv [46] utilize a linear combination of parallel static kernels, with the scaling factors dynamically aggregated according to individual inputs. The dynamic filter network [47] generates spatial-variant kernels but tends to be computationally intensive. To address this, decoupled dynamic filter (DDF) [48] decouples a depth-wise dynamic filter into spatial and channel dynamic filters, which reduces parameter counts and computational costs significantly. MGDN [49] demonstrates that generating spatial-variant kernels for different locations, guided by additional input data, can enhance performance in multimodal image fusion tasks. In contrast, the proposed CMDF module extends the principles of DDF to multimodal object detection, innovatively designing convolutions that blend dynamic and static properties to effectively mitigate modality redundancy.

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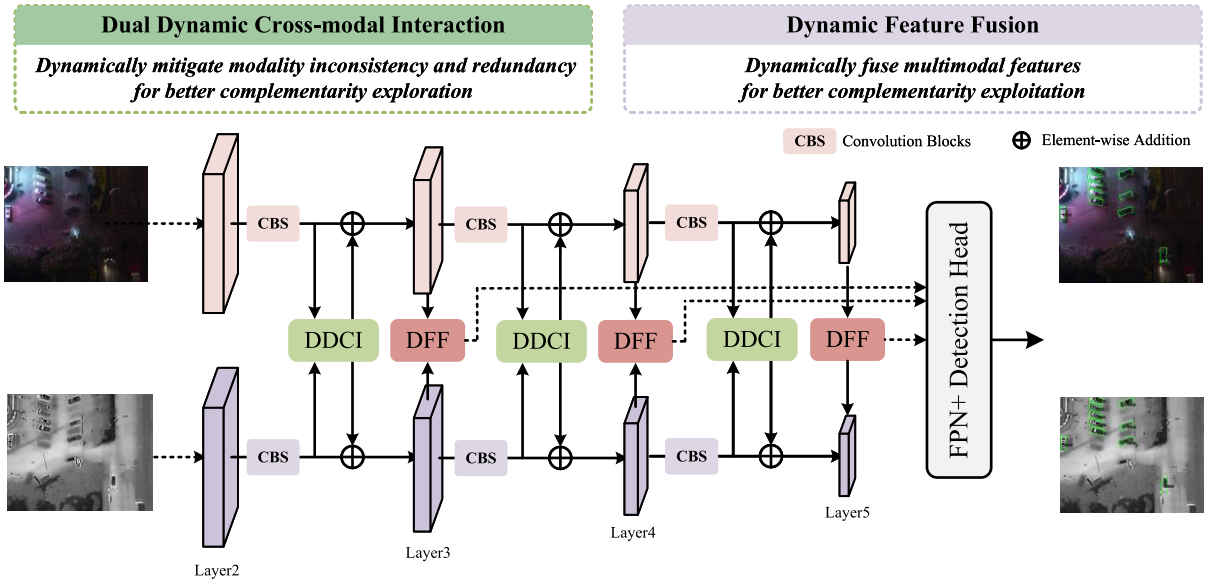


Fig. 2. Overall framework for a DDCINet consists of the DDCI module and the DFF module, positioned after the last three feature extraction layers in a two-stream backbone. The DDCI module dynamically mitigates modality inconsistency and redundancy, facilitating better complementarity exploration. Meanwhile, the DFF module dynamically fuses multimodal features to enhance complementarity exploitation. CBS represents the convolution blocks of the optical and infrared branches.

III. Methodology

In this section, we first present the motivation behind our proposed DDCINet and then introduce the overview of the DDCINet. Next, the design of the DDCI module and the DFF module is presented in detail.

A. Motivation

Modality inconsistency and redundancy significantly hinder the effective extraction of complementary features, leading to suboptimal feature fusion. Thus, a key factor in achieving efficient MM-RSOD is the implementation of better extraction methods for complementary features to facilitate subsequent feature fusion. Due to the different imaging mechanisms of multimodal images, features from one modality can enhance the extraction of complementary features from another modality. Therefore, this article proposes the DDCI module to dynamically mitigate modality inconsistency and redundancy, enabling better complementarity exploration through cross-modal interaction. After complementary feature extraction, semantic-spatial misalignment and modality-specific uncertainty are reduced. To further avoid the suboptimality caused by direct feature fusion via simple addition, this article proposes the DFF module, which dynamically fuses features for improved complementarity exploitation, maximizing the potential of complementary feature extraction.

B. Overview

The overview of the DDCINet for MM-RSOD is depicted in Fig. 2. The DDCINet utilizes two-stream backbones to extract 5-layer features from input images. The DDCI module is integrated into the last three layers to dynamically and simultaneously address modality inconsistency and redundancy. While

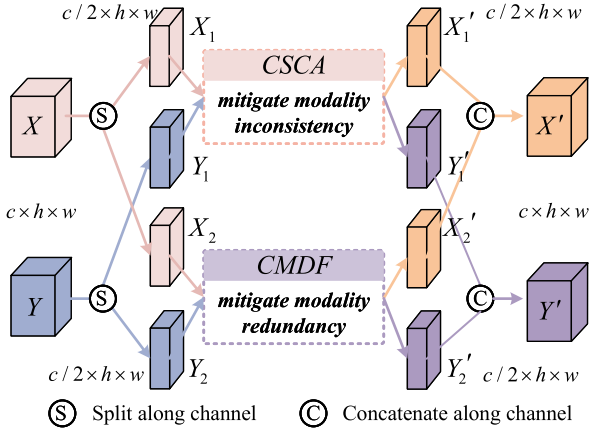


Fig. 3. Overall workflow for the DDCI module simultaneously mitigates modality inconsistency and redundancy by collaboratively applying the CSCA and CMDF on evenly split multimodal feature segments. This "split-interact-fuse" design enhances the network's capability for better complementarity exploration.

DDCI can be applied at each level, we focus on the last three layers to balance speed and accuracy (for a detailed analysis, refer to Section IV-D3). This integration ensures that the enhanced features are forwarded to subsequent stages, further deepening and improving complementary feature extraction. Additionally, the DFF module dynamically fuses the multimodal features from these layers into a single representation. Finally, a detection head (note that the FPN [50] structure is omitted in this framework) is utilized for the final detection task.

C. Dual-Dynamic Cross-Modal Interaction

The DDCI module is inspired by the feature partition operation [43], [51] and performs dual-cross-modal interaction

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on evenly split multimodal feature segments to dynamically and simultaneously mitigate modality inconsistency and redundancy. The overall workflow of the DDCI module is illustrated in Fig. 3. Assuming that optical and infrared (SAR) features are denoted by $X \in \mathbb{R}^{C\times H\times W}$ and $Y \in \mathbb{R}^{C\times H\times W}$, respectively, the overall workflow of the DDCI module can be mathematically formulated as

$$\begin{aligned} X_1, X_2 &= \text{Split}\big(\text{Conv}_{1\times1}^{C\to C}(X)\big) \\ Y_1, Y_2 &= \text{Split}\big(\text{Conv}_{1\times1}^{C\to C}(Y)\big) \\ X_1', Y_1' &= \text{CSCA}(X_1, Y_1) \\ X_2', Y_2' &= \text{CMDF}(X_2, Y_2) \\ X' &= \text{Conv}_{1\times1}^{C\to C}\big(\text{Cat}\big(X_1', X_2'\big)\big) \\ Y' &= \text{Conv}_{1\times1}^{C\to C}\big(\text{Cat}\big(Y_1', Y_2'\big)\big). \end{aligned} \tag{1}$$

The term $\text{Conv}_{1\times1}^{C\to C}$ represents a standard convolution operation with a kernel size of 1×1, used to convert the number of channels from C to C. Split is the split operation along the channel dimension, while Cat is the concatenation operation along the channel dimension. To address the modal consistency and redundancy while increasing as few parameters as possible, the DDCI module splits X into $X_1, X_2 \in \mathbb{R}^{(C/2)\times H\times W}$ uniformly along the channel dimension. Similarly, Y is also uniformly divided into $Y_1, Y_2 \in \mathbb{R}^{(C/2)\times H\times W}$. Then, X_1 and Y_1 are processed through the CSCA module to address modality inconsistency, whereas X_2 and Y_2 undergo processing in the CMDF module to alleviate modality redundancy. More specifically, the CSCA component enhances the semantic-spatial correlation between modalities by identifying the most relevant channel–spatial features through cross-attention, addressing modality inconsistency (see Section III-C1 for more detailed introduction). The CMDF component achieves cross-modal context interaction through static convolution and further generates dynamic spatial-variant kernels to filter out irrelevant information between modalities, addressing modality redundancy (see Section III-C2 for more detailed introduction). The output features $\{X_1', X_2'\} \in \mathbb{R}^{(C/2)\times H\times W}$ and $\{Y_1', Y_2'\} \in \mathbb{R}^{(C/2)\times H\times W}$ are then concatenated along the channel dimension to generate new two-modal feature maps $X' \in \mathbb{R}^{C\times H\times W}$ and $Y' \in \mathbb{R}^{C\times H\times W}$. Additionally, a single-layer network is incorporated before the split and after the concatenation operations to enhance representation learning and collaborative capabilities. Stacking DDCI modules enables dynamic interactions between multimodal features, effectively addressing modality inconsistency and redundancy, and encapsulating robust complementarity exploration and exploitation capabilities within the model.

1) Channel-Gated Spatial Cross-Attention: CSCA enhances the semantic-spatial correlation between modalities by identifying the most relevant channel–spatial features through cross-attention. Utilizing cross-attention [11] to establish relationships between multimodal features and address modality inconsistency has been demonstrated in several works [19], [20], [21], [22]. Cross-attention achieves modality complementarity by identifying the most relevant component between each query position and all other positions globally. Due to the significant computational and memory costs, basic spatial cross-attention is impractical for high-resolution feature maps. Thus, these methods aggregate local information via spatial reduction (SR) operation before calculating similarities. However, SR simply adopts linear projection to reduce input dimension but ignores efficient aggregation of local information. In response, we design the multiscale SR (MSSR) to assist multihead cross-attention in efficiently aggregating local information and exploring multiscale complementarity. Moreover, spatial cross-attention focuses on spatial relationships but ignores channel-wise complementarity. Differently, we design the channel-gated feed-forward network (CG-FFN) to augment channel relationships after spatial cross-attention. CG-FFN is inspired by the gated linear unit (GLU) [52] and squeeze-and-excitation (SE) mechanism [37], which use cross-channel attention scores as the gating function to further enhance FFN features for more comprehensive complementarity exploration. The overall workflow of CSCA is illustrated in Fig. 4 and can be mathematically formulated as follows (for convenience, we only show the processing for modality X_1, which is consistent with modality Y_1):

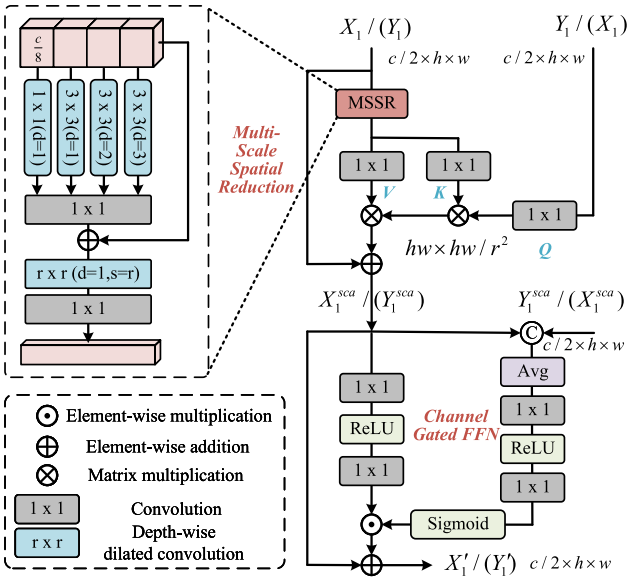


Fig. 4. Overall workflow for the CSCA module enhances the semantic-spatial correlation between modalities by identifying the most relevant channel–spatial features through cross-attention, addressing modality inconsistency. The MSSR efficiently aggregates local information and explores multiscale complementarity. Meanwhile, the CG-FFN augments channel relationships following the spatial cross-attention process.

$$\begin{aligned} Q_Y &= \text{Reshape}_1\big(\text{Conv}_{1\times1}^{\frac{C}{2}\to\frac{C}{2}}(Y_1)\big) \\ K_X, V_X &= \text{Split}\big(\text{Reshape}_1\big(\text{Conv}_{1\times1}^{\frac{C}{2}\to C}(\text{MSSR}(X_1))\big)\big) \\ X_1^{\text{sca}} &= \text{Conv}_{1\times1}^{\frac{C}{2}\to\frac{C}{2}}\left(\text{Reshape}_2\left(\text{softmax}\left(\frac{Q_Y K_X^T}{\sqrt{d}}\right)V_X\right)\right) \\ Z &= \text{Cat}(\text{X}_1^{\text{sca}}\|\text{Y}_1^{\text{sca}}) \\ g &= \text{Sig}\big(\text{Conv}_{1\times1}^{\frac{C}{r}\to\frac{C}{2}}\big(\text{ReLU}\big(\text{Conv}_{1\times1}^{C\to\frac{C}{r}}(\text{Avg}(\text{Z}))\big)\big)\big) \\ X_1' &= \text{Conv}_{1\times1}^{C\to\frac{C}{2}}\big(\text{ReLU}\big(\text{Conv}_{1\times1}^{\frac{C}{2}\to C}(X_1^{\text{sca}})\big)\big)\cdot g. \end{aligned} \tag{2}$$

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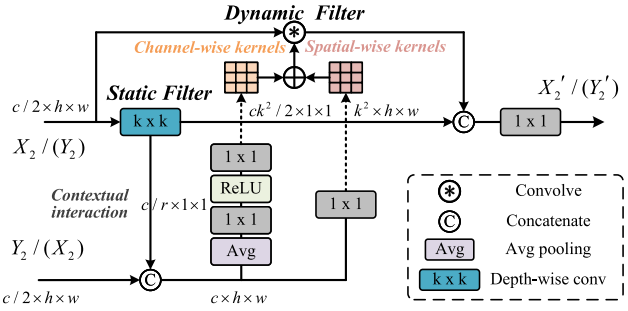


Fig. 5. Overall workflow for the CMDF module achieves cross-modal context interaction through static convolution and further generates dynamic spatial-variant kernels to filter out irrelevant information between modalities, addressing modality redundancy.

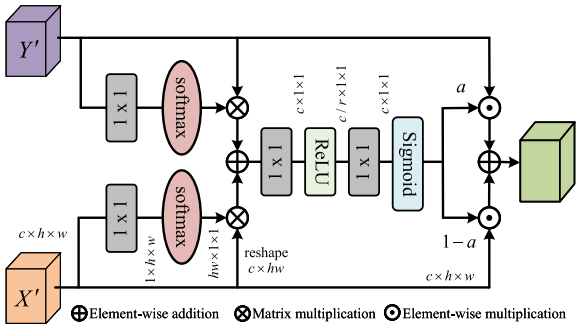


Fig. 6. Overall workflow for the DFF module dynamically adjusts interchannel dependencies guided by modal-specific global context to fuse features, achieving better complementarity exploitation.

Here, Reshape_1 refers to reshaping the feature from $C \times H \times W$ to a sequence with $HW \times C$ dimensions, while Reshape_2 is its inverse operation. Sig and ReLU denote the sigmoid and ReLU activation functions, respectively. $(\cdot)$ signifies element-wise multiplication with broadcast. d represents the number of channels in each attention head. In the MSSR block illustrated in Fig. 4, we divide X_1 into four segments along the channel dimension and apply depth-wise dilated convolution with varying dilation rates to extract multiscale features. These features are then fused using a convolutional layer. MSSR features can be obtained using depth-wise separable convolution [53] with a kernel size of $s \times s$. Here, s is a hyperparameter that balances the network's accuracy and complexity. Next, K_X and V_X extracted from MSSR, along with Q_Y extracted from Y_1, will be input to the multihead cross-attention module [11] to produce X_1^{sca}. The CG-FFN comprises two branches: the FFN branch and the cross-channel gated branch. The FFN branch consists of two linear projections and is primarily used for channel mixing within the modality. On the other hand, the cross-channel gated branch adopts the SE [37] block to obtain channel attention scores g from multimodal features. Finally, the outputs of the FFN branch and the gated branch are multiplied to enhance channel-wise complementarity. In summary, the CSCA module can leverage both spatial and channel cross-attention to establish effective cross-modal relationships, mitigating modality inconsistency.

2) Cross-Modal Dynamic Filters: The information from different modalities can be both complementary and redundant. Traditional static filters slide uniformly over images, treating all pixels equally, which overlooks the spatial variance of noise in different locations. Moreover, static filters focus on a single input, failing to leverage useful data from multimodal sources. This limitation results in inadequate preservation of complementary information and ineffective suppression of irrelevant data. To address these challenges, we draw inspiration from DDF [48] and MGDN [49], introducing the CMDF module. This module achieves cross-modal context interaction using static convolution while generating dynamic spatial-variant kernels to filter out irrelevant information between modalities. The overall workflow of the CMDF module is illustrated in Fig. 5 and can be mathematically formulated as follows (for clarity, we showcase the process for modality X_2, aligning with modality Y_2):

$$
\begin{aligned}
X_2^{\text{static}} &= \text{DWConv}_{k\times k}^{\frac{C}{2}\to\frac{C}{2}}(X_2) \\
F &= \text{Cat}\left(\text{X}_2^{\text{static}}, \text{Y}_2\right) \\
K_{\text{spatial}} &= \text{Conv}_{1\times 1}^{C\to k^2}(F) \\
K_{\text{channel}} &= \text{Conv}_{1\times 1}^{\frac{C}{r}\to\frac{Ck^2}{2}}\left(\text{ReLU}\left(\text{Conv}_{1\times 1}^{C\to\frac{C}{r}}(\text{Avg(F)})\right)\right) \\
X_2^{\text{dynamic}} &= \mathcal{F}(X_2; K_{\text{spatial}} \oplus K_{\text{channel}}) \\
X_2' &= \text{Conv}_{1\times 1}^{C\to\frac{C}{2}}\left(\text{Cat}\left(\text{X}_2^{\text{static}}, \text{X}_2^{\text{dynamic}}\right)\right)
\end{aligned} \tag{3}
$$

where DWConv represents the depth-wise convolution, $\mathcal{F}$ denotes the convolution operation, Avg signifies spatial-wise average pooling, and r indicates the ratio of channel compression. Initially, the CMDF module applies a static depth-wise filter to process spatially invariant noisy information across the feature map, generating the static features X_2^{static}. Subsequently, the module concatenates X_2^{static} with modality Y_2 to derive cross-modal contextual features F. The use of X_2^{static} as a static input representation reflects contextual information among local neighbors, facilitating straightforward cross-modal contextual interaction via concatenation. These contextual features F undergo processing in the spatial-variant filter generation block to naturally exploit mutual relations between the two modalities with the guidance of static context. Following DDF [48], the spatial-variant filter generation is decomposed into spatial-wise and channel-wise filter generation to minimize parameters and computational costs. Here, $K_{\text{spatial}} \in \mathbb{R}^{k^2\times h\times w}$ and $K_{\text{channel}} \in \mathbb{R}^{(Ck^2/2)\times 1\times 1}$ represent the generated spatial-wise and channel-wise filters, respectively. Then, the overall dynamic filters $K \in \mathbb{R}^{(C/2)\times k\times k\times h\times w} = K_{\text{spatial}} \oplus K_{\text{channel}}$ are then convolved across X_2 to filter spatial-variant noisy information, yielding dynamic features X_2^{dynamic}. $\oplus$ signifies element-wise addition with broadcast. Finally, the concatenation of X_2^{static} and X_2^{dynamic} along the channel dimension followed by simple fusion layers yields the enhanced complementary features X_2'.

D. Dynamic Feature Fusion

After complementary feature extraction, semantic-spatial misalignment and modality-specific uncertainty are mitigated. To further avoid the suboptimality caused by the direct

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feature fusion via simple addition operation, the DFF module dynamically adjusts interchannel dependencies guided by modal-specific global context, achieving superior feature fusion and better complementarity exploitation. The DFF operation is illustrated in Fig. 6 and can be mathematically formulated as

$$\begin{aligned} G_x &= \text{Reshape}_3\left(\text{softmax}\left(\text{Conv}_{1\times1}^{C\to1}(X')\right)\right) \\ G_y &= \text{Reshape}_3\left(\text{softmax}\left(\text{Conv}_{1\times1}^{C\to1}(Y')\right)\right) \\ S &= \text{Reshape}_4(X')G_x + \text{Reshape}_4(Y')G_y \\ a &= \text{Sigmoid}\left(\text{Conv}_{1\times1}^{\frac{C}{r}\to C}\left(\text{ReLU}\left(\text{Conv}_{1\times1}^{C\to\frac{C}{r}}(S)\right)\right)\right) \\ Z &= a\cdot X' + (1-a)\cdot Y' \end{aligned} \tag{4}$$

where Reshape_3 reshapes the feature from the shape of $1 \times H \times W$ to $HW \times 1 \times 1$, while Reshape_4 reshapes the image feature from $C \times H \times W$ to $C \times HW$ as a sequence. The DFF process integrates multimodal features by modeling complementary channel dependencies guided by modality-specific global complementarity. Initially, we extract modality-specific global contexts $G_x \in \mathbb{R}^{HW\times1\times1}$ and $G_y \in \mathbb{R}^{HW\times1\times1}$ through a convolution layer followed by a softmax activation and reshaping operation. Subsequently, the modality-specific global context is multiplied with the respective modality features across the channel dimension and then aggregated to obtain the fused channel response $S \in \mathbb{R}^{C\times1\times1}$. Passing S through two consecutive convolution layers yields the fusion factor a, effectively capturing interdependencies between channels. Finally, this fusion factor guides the multimodal feature fusion process.

IV. Experiments

In this section, we begin by detailing the datasets and evaluation metrics utilized in our experiments. Following that, we present a comprehensive set of quantitative results, which includes comparisons with closely related detectors, ablation studies, and generalizability. Additionally, we provide visualized results to enhance understanding and illustrate our findings more intuitively.

A. Datasets

As mentioned in Section I, MM-RSOD usually involves optical, infrared, and SAR modalities. Therefore, we conduct experiments on DroneVehicle [27] and VEDAI [28] datasets for optical–infrared object detection tasks, because they are both widely used benchmarks to evaluate optical–infrared object detection methods [18], [20], [26]. As for the optical–SAR object detection task, we conduct experiments on an extremely representative and challenging dataset: OGSOD [29].

DroneVehicle [27] dataset is a large-scale drone-based optical–infrared dataset that contains 28 439 image pairs and 953 087 vehicle instances. The dataset covers multiple scenarios, including urban roads, residential areas, parking lots, and other scenarios from day to night. Rich annotations have been made by the authors with oriented bounding boxes for the five categories (cars, buses, trucks, vans, and freight cars). We remove the white border of the images to maintain an image size of 640 × 512.

VEDAI [28] dataset is a public dataset for optical–infrared object detection in aerial imagery, containing more than 3700 annotated targets in 1268 image pairs. We use eight vehicle categories in this dataset, including cars, pickup, camping cars, trucks, other, tractors, boats, and vans. We use images with a size of 512 × 512 for training and testing. The rotated bounding box annotation format is used to evaluate our proposed methods.

OGSOD [29] dataset is a recently released optical–SAR paired dataset for MM-RSOD. It consists of a training set of 14 665 image pairs and a test set of 3666 image pairs, containing over 48 000 instance annotations in total. The image size for all images is 256 × 256. Three categories, including bridges, storage tanks, and harbors, are annotated, and the rotated bounding box annotation format is used to evaluate our proposed methods.

B. Evaluation Metrics

We employ mean average precision (mAP) to evaluate the performance of object detectors. Intersection over union (IoU) threshold is utilized to determine true positives (TPs) and false positives (FPs). For a more detailed evaluation, we set the IoU threshold as 0.5 and 0.75, forming mAP50 and mAP75, respectively.

C. Implementation Details

$\mathbf{S}^2$A-Net [34], RoI Transformers [35], and Oriented R-CNN [36] are representative RSOD methods, while ResNet50 [54] and Swin-Transformer-T [42] are widely used backbones. Therefore, we use these three methods with two backbones as the basic benchmarks to perform our proposed DDCINet in this article. All experiments are conducted using MMRotate (https://github.com/open-mmlab/mmrotate) as the base framework. The pretrained ResNet50 [54] on ImageNet [55] is adopted to initialize the two-stream backbone network. All models are trained using stochastic gradient descent (SGD) for 12 epochs (1× schedule training strategy) with 16 images per batch on an NVIDIA V100 GPU. We set the learning rate to 0.02 for RoI Transformers and Oriented R-CNN and 0.01 for $\mathbf{S}^2$A-Net, then divide it by 10 at the 8th and 11th epochs. Other parameters are set identical to those in MMRotate. For data augmentation, we only apply random horizontal and vertical flips with a probability of 0.5. During testing, the IoU threshold is set to 0.1 for rigorous filtering of bounding boxes with low precision.

D. Results Analysis

1) Overall Performance: We evaluated the DDCINet against several SOTA detectors on the DroneVehicle test dataset [27] to demonstrate the superiority of our proposed method, as shown in Table I. Specifically, we integrated the DDCINet into $\mathbf{S}^2$A-Net [34] and RoI Transformer [35] benchmarks, creating new oriented object detectors named DDCINet-$\mathbf{S}^2$A-Net and DDCINet-RoI-Trans, respectively. The

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TABLE I

COMPREHENSIVE EXPERIMENTS ON THE DRONEVEHICLE TEST DATASET [27]. WE COMPARE OUR DDCINET BASED ON $\mathbf{S}^2$A-NET [34] AND ORIENTED R-CNN [36] BENCHMARK WITH SEVERAL SINGLE-MODAL AND MULTIMODAL DETECTORS. THE BOLD IS THE BEST AND THE UNDERLINE REPRESENTS THE SECOND-BEST. WE TRAIN OUR MODELS WITH 12 EPOCHS BY DEFAULT AND * REPRESENTS THAT MODEL IS TRAINED WITH 20 EPOCHS

Detectors	Modality	Backbone	Type	Car	Truck	Freight car	Bus	Van	mAP50
$\mathbf{S}^2$A-Net [34]	Optical	ResNet50	One-stage	79.9	50.0	36.2	82.8	37.5	57.3
RoI Transformer [35]		ResNet50	Two-stage	61.6	55.1	42.2	85.5	44.8	61.6
Oriented R-CNN [36]		ResNet50	Two-stage	80.3	55.4	42.1	86.8	46.9	62.3
$\mathbf{S}^2$A-Net [34]	Infrared	ResNet50	One-stage	89.7	51.0	50.2	89.0	44.0	64.8
RoI Transformer [35]		ResNet50	Two-stage	89.6	51.0	53.4	88.9	44.5	65.5
Oriented R-CNN [36]		ResNet50	Two-stage	89.6	53.9	53.9	89.2	41.0	65.5
CMFR [23]	Optical + Infrared	ResNet50	One-stage	90.1	64.4	62.4	88.8	53.6	72.9
CALNet [21]		ResNet50	One-stage	90.2	73.8	60.9	88.7	51.7	73.1
$\mathbf{C}^2$Former [20]		ResNet50	One-stage	90.2	68.3	64.4	89.8	58.5	74.2
DDCINet-$\mathbf{S}^2$A-Net (ours)		ResNet50	One-stage	90.4	75.5	62.0	89.7	60.9	75.7
UA-CMDet [27]		ResNet50	Two-stage	88.6	73.0	56.0	88.3	54.8	72.2
TSFADet [17]		ResNet50	Two-stage	89.9	67.9	63.7	89.8	54.0	73.1
CLGNet [26]		ResNet50	Two-stage	90.4	72.6	64.6	89.9	59.5	75.4
CAGT [18]		ResNet50	Two-stage	<u>90.9</u>	69.7	**66.3**	<u>90.5</u>	55.6	74.6
DATFF [22]		ResNet50	Two-stage	90.4	78.1	63.6	89.9	58.1	76.0
DDCINet-RoI-Trans (ours)		ResNet50	Two-stage	90.5	<u>78.6</u>	64.9	90.0	<u>64.9</u>	<u>77.5</u>
DDCINet-RoI-Trans* (ours)		ResNet50	Two-stage	**91.0**	**78.9**	<u>66.1</u>	**90.7**	**65.5**	**78.4**

TABLE II

ABLATION STUDY FOR THE DDCINET ON DRONEVEHICLE [27] VALIDATION DATASET AND $\mathbf{S}^2$A-NET [34] BENCHMARK

DDCINet			mAP50	mAP75	Params(M)	FLOPs(G)	FPS
DDCI		DFF					
CSCA	CMDF						
			71.0	49.5	**59.48**	**88.35**	**24.8**
✓			74.9	54.3	108.42	120.65	17.8
	✓		73.6	54.0	94.77	108.67	20.0
✓	✓		75.2	55.2	120.30	124.85	17.3
		✓	71.6	50.5	60.18	88.36	24.7
✓	✓	✓	**75.7**	56.4	121.70	124.86	17.2

TABLE III

ABLATION STUDY FOR THE CSCA MODULE ON THE DRONEVEHICLE [27] VALIDATION DATASET AND $\mathbf{S}^2$A-NET [34] BENCHMARK

CSCA				mAP50	mAP75	Params(M)	FLOPs(G)
SR-Spa	FFN	MSSR	CG				
				71.0	49.5	**59.48**	**88.35**
✓				73.8	53.1	93.22	110.50
✓	✓			74.0	53.4	104.25	118.58
✓	✓	✓		74.4	54.0	107.04	120.64
✓	✓		✓	74.6	53.8	105.63	118.59
✓	✓	✓	✓	**74.9**	54.3	108.42	120.65

compared single-modal detectors include $\mathbf{S}^2$A-Net [34], RoI Transformer [35], and Oriented R-CNN [36]. Regarding the multimodal detectors, we compared UA-CMDet [27], TSFADet [17], CAGT [18], DATFF [22], CLGNet [26], CMFR [23], CALNet [21], and $\mathbf{C}^2$Former [20]. It is evident that the results of multimodal detectors outperform those of single-modal detectors, highlighting the complementarity between multimodal remote-sensing images. Among the multimodal detectors, the DDCINet-$\mathbf{S}^2$A-Net and DDCINet-RoI-Trans achieve 75.7% and 77.9% mAP50, outperforming all other one-stage and two-stage methods, respectively. In particular, the DDCINet-$\mathbf{S}^2$A-Net and DDCINet-RoI-Trans achieve detection accuracy 1.5% and 1.9% higher than the corresponding second-best methods ($\mathbf{C}^2$Former [20] and DATFF [22]), respectively. Because some methods, such as $\mathbf{C}^2$Former [20] and CAGT [18], use a greater number of

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TABLE IV

ABLATION STUDY FOR THE CMDF MODULE ON THE DRONEVEHICLE [27] VALIDATION DATASET AND S^2A-NET [34] BENCHMARK

CMDF			mAP50	mAP75	Params(M)	FLOPs(G)
Static	Spa	Chn				
			71.0	49.5	**59.48**	**88.35**
✓			71.3	49.7	84.31	106.54
	✓		71.9	52.0	84.34	106.58
		✓	71.7	49.9	88.12	106.51
	✓	✓	73.1	53.2	91.97	106.60
✓	✓	✓	**73.6**	54.0	94.77	108.67

TABLE V

ABLATION STUDY FOR ADDING THE DDCI MODULE TO DIFFERENT STAGES ON THE DRONEVEHICLE [27] VALIDATION DATASET AND S^2A-NET [34] BENCHMARK

different stages for DDCI module				mAP50	Params(M)	FLOPs(G)
stage2	stage3	stage4	stage5			
				71.0	**59.48**	**88.35**
			✓	74.7	106.32	101.15
		✓	✓	75.3	118.61	113.15
	✓	✓	✓	75.7	121.70	124.86
✓	✓	✓	✓	**75.8**	122.51	136.64

training epochs to enhance performance, we additionally train the DDCINet-RoI-Trans* with 20 epochs for the fair comparison. It can be observed that the DDCINet-RoI-Trans* can further improve mAP50 by 0.9% and achieves the best performance in all categories except for the freight car category, where comparable performance is also attained. Specifically, CAGT achieves 66.3% AP50 for freight cars, while the DDCINet-RoI-Trans* achieves 66.1% AP50 for freight cars. These results comprehensively demonstrate the ability of the DDCINet to mitigate modality inconsistency and redundancy via DDCI.

2) Ablation Study: We conducted a series of ablation studies to analyze the influence of each component designed in the DDCINet, the CSCA module, and the CMDF module based on the DroneVehicle validation dataset and the S^2A-Net benchmark.

a) Ablation study for the DDCINet: Table II presents the detection results of each component in the DDCINet. The CSCA block and the CMDF block improve the mAP50 by 3.9% and 2.6%, respectively, compared to the baseline. Correspondingly, the parameters and FLOPs of both blocks also increase, resulting in a decrease in FPS of 6 and 4.8, respectively. Moreover, as the IoU threshold increases, indicating a higher requirement for localization accuracy, mAP75 also increases significantly. When simultaneously adopting the CSCA block and the CMDF block, mAP50 and mAP75 continue to improve, achieving 75.2% and 55.2%, respectively. These performance improvements illustrate that the CSCA and CMDF blocks can effectively mitigate modality inconsistency and redundancy. The DFF module boosts the performance by 0.6% and 0.5% in terms of mAP50 compared to the baseline and the baseline with the DDCI module, respectively, verifying DDCINet's ability to explore and exploit complementary features for MM-RSOD. Next, we will conduct more detailed ablation studies on the CSCA and CMDF modules to verify the effectiveness of each design choice.

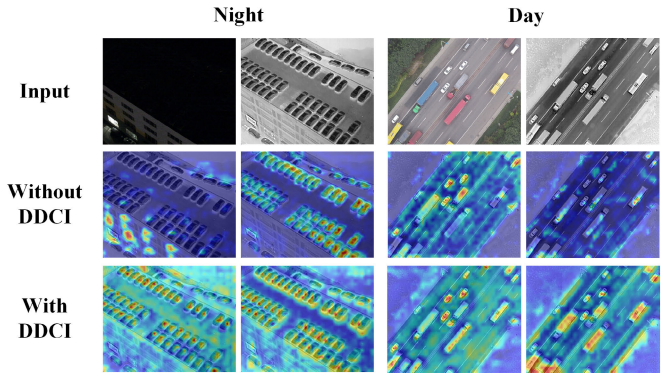


Fig. 7. Visualization of feature maps after Layer3 without and with the DDCI module.

b) Ablation study for the CSCA module: Table III presents the detection results of each component in the CSCA module. When only adopting spatial-reduction cross-attention, the SR-Spa method improves mAP50 performance by 2.8% with an increase of 22.15G FLOPs compared to the baseline. After adding the FFN, mAP50 and mAP75 can further improve by 0.2% and 0.3%, respectively. Due to efficient multiscale information aggregation and complementarity exploration, MSSR can increase mAP50 by 0.4% compared to SR-Spa-FFN. By using the Channel-Gated FFN, mAP50 can be improved by 0.6% because the cross-channel mixer augments channel cross-modal relationships. Moreover, both the MSSR and CG-FFN are lightweight architectures with only a few additional parameters and FLOPs, verifying the effectiveness of each component and the CSCA module's ability to mitigate modality inconsistency.

c) Ablation study for the CMDF module: Table IV presents the detection results of each component in the CMDF module. Using only the static filter brings only a 0.3% mAP50 improvement compared to the baseline. The dynamic filter is decoupled into spatial-wise and channel-wise dynamic filters, which improve mAP50 performance by 0.9% and 0.7%, respectively. Specifically, the spatial-wise dynamic filter significantly boosts mAP75 performance by 2.5%, demonstrating that leveraging multimodal information to generate spatial-variant kernels is conducive to preserving complementary information and suppressing irrelevant information in the MM-RSOD task. Consequently, the dynamic filter achieves a 73.1% mAP50 performance with only a negligible increase in FLOPs compared to the static filter. Finally, when combining the dynamic filter and static filter, our proposed CMDF module significantly improves accuracy, with a 2.6% mAP50 improvement and a 4.5% mAP75 improvement. The effectiveness of each module and CMDF's ability to mitigate modality redundancy are verified.

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TABLE VI
OVERALL PERFORMANCE ON DIFFERENT BACKBONES, BENCHMARKS AND DATASETS

Benchmarks	Backbone	Method	DroneVehicle [27]		VEDAI [28]		OGSOD [29]	
			mAP50	mAP75	mAP50	mAP75	mAP50	mAP75
RoI Transformer [35]	ResNet50	baseline	73.6	58.7	74.1	33.4	79.6	36.7
		DDCINet	**77.5** (3.9↑)	**63.2** (4.5↑)	**75.1** (1.0↑)	**39.2** (5.8↑)	**80.8** (1.2↑)	**39.1** (2.4↑)
Oriented R-CNN [36]	ResNet50	baseline	73.0	56.1	69.1	25.9	73.9	29.9
		DDCINet	**77.4** (4.4↑)	**60.9** (4.8↑)	**73.5** (4.4↑)	**31.0** (5.1↑)	**77.0** (3.1↑)	**34.5** (4.6↑)
Oriented R-CNN [36]	Swin-T	baseline	75.8	57.5	75.8	35.4	79.3	36.3
		DDCINet	**77.4** (1.6↑)	**58.9** (1.4↑)	**77.0** (1.2↑)	**38.6** (3.2↑)	**81.1** (1.8↑)	**40.1** (3.8↑)

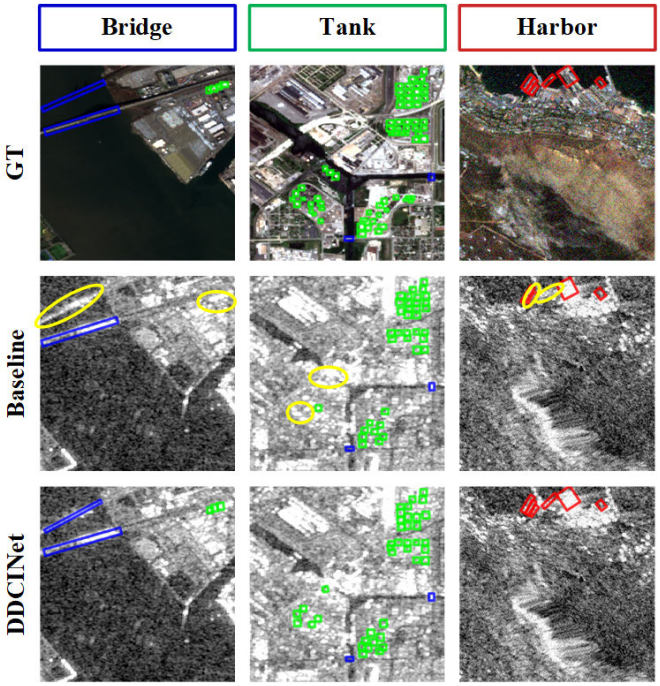


Fig. 8. Visualization of some detection results of different methods in the OGSOD test dataset [29]. The confidence threshold is set to 0.5 when visualizing these results. The orange circles represent the missing detection.

3) Different Stages for the DDCI Module: We added the proposed DDCI module to the baseline network (two-stream S^2A-Net) at different stages. Observing Table V, we can note that gradually adding the DDCI module from stage 5 to stage 2 improves detection performance accordingly, along with an increase in parameters and FLOPs. Ultimately, the DDCINet achieves 75.7% mAP50, 121.70M parameters, and 124.85G FLOPs after adding DDCI in the last three stages. However, continuing to add DDCI to stage 2 only improves mAP50 accuracy by 0.2% with an 11.79G FLOPs increase instead. Hence, we added DDCI in the last three stages of all experiments to achieve a balance between speed and accuracy.

4) Generalizability: We integrate our DDCINet into different backbones and benchmarks to form various detectors and conduct experiments on different datasets to verify the generalizability of our method. We select 2 backbones including ResNet50 [54] and Swin-Transformer-T [42], two benchmarks including RoI Transformer [35] and Oriented R-CNN [36], and three MM-RSOD datasets including DroneVehicle [27], VEDAI [28], and OGSOD [29]. The results are provided in Table VI, and we can see that our DDCINet can improve both mAP50 and mAP75 compared to all baseline methods on all backbones, benchmarks, and datasets. Specifically, when adopting ResNet50 as the backbone, our DDCINet can improve mAP50 and mAP75 by 4.4% and 4.8%, respectively, for the Oriented R-CNN benchmark on the DroneVehicle dataset. The quantitative improvement in detection results significantly demonstrates the effectiveness and generality of our DDCINet.

5) Visualization Results: In addition to quantitative comparisons, we also visualize feature maps and detection results on the DroneVehicle validation dataset in Figs. 7–9 to provide an intuitive understanding of our proposed methods.

a) Visualization of feature maps: Fig. 7 presents the intermediate feature maps of multimodal detectors without and with the DDCI module. In nighttime conditions, optical images exhibit significant uncertainty due to low-light conditions, limiting both object and background feature extraction. Guided by the DDCI module, object features are emphasized while background features are suppressed, leading to obvious complementarity for optical and infrared modalities. For daytime scenarios, optical images capture different vehicle types well, while the infrared modality recognizes only a few vehicles. The proposed DDCI module effectively alleviates this semantic misalignment and increases the prominence of objects with different categories, demonstrating the effectiveness of the DDCI module.

b) Visualization of detection results in optical–SAR imagery: Fig. 8 displays the visual detection results in the OGSOD test dataset [29]. From top to bottom, we have GroundTruth (optical modality), Baseline [36], and DDCINet, respectively. It is obvious that the baseline misses several bridges, tanks, and harbors due to inadequate extraction of complementary features. In contrast, the DDCINet significantly improves the detection results compared to the baseline. This verifies the significant generalizability of our proposed method on multimodal object detection imagery.

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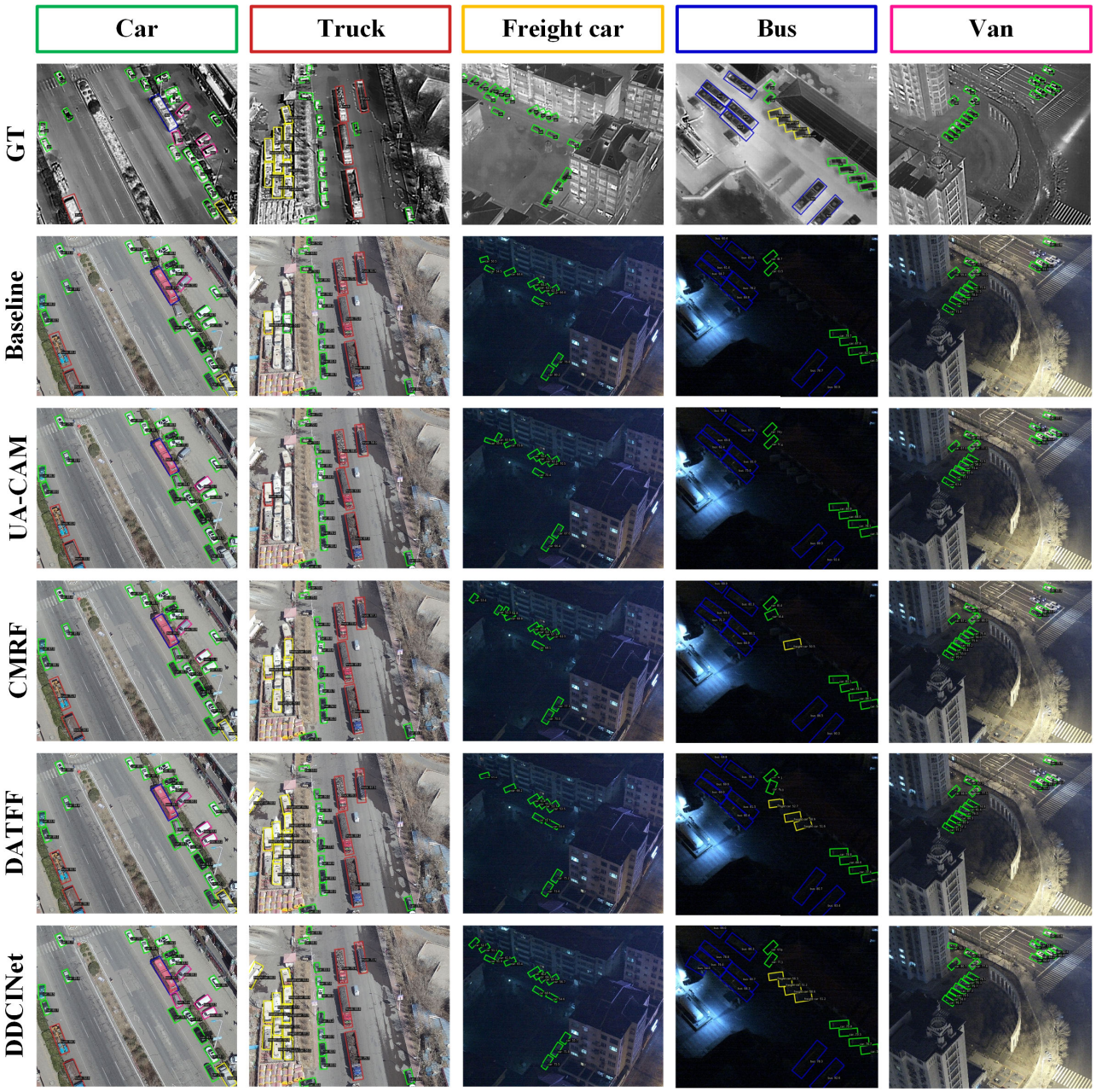


Fig. 9. Visualization of some detection results of different methods in the DroneVehicle validation dataset [27]. The confidence threshold is set to 0.5 when visualizing these results.

c) Visualization of detection results in optical–infrared imagery: Fig. 9 presents the visual detection results of several methods in the DroneVehicle test dataset [27]. From top to bottom, we have GroundTruth (infrared modality), Baseline [34], UA-CMDet [27], CMFR [23], DATFF [22], and DDCINet. Due to extreme modality inconsistency and redundancy, results from other methods exhibit misclassification between car and van, along with a number of missed detections for cars and freight cars. The DDCI module aims to mitigate modality inconsistency and redundancy, enabling the DDCINet to achieve better classification and more TPs. Despite its superior performance compared to other methods, the DDCINet still has room for improvement in detecting objects in complex scenes, which should be addressed in future work.

V. Conclusion

This article addresses two critical challenges in MM-RSOD: modality inconsistency and redundancy, which hinder the effective extraction and fusion of complementary features. Modality inconsistency can cause semantic-spatial misalignment, while redundancy introduces modality-specific uncertainty. To tackle these issues, we propose the DDCINet, which includes the DDCI module and the DFF module for enhanced complementarity exploration and exploitation. The DDCI module mitigates both modality inconsistency and redundancy by employing the CSCA and CMDF blocks collaboratively on evenly split multimodal feature segments. This approach enhances the exploration of complementary features. Meanwhile, the DFF module dynamically adjusts

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interchannel dependencies, guided by modal-specific global context, to improve feature fusion and complementarity exploitation. We conducted various experiments to validate the superiority and generalizability of the DDCINet. Our work presents a unified, dynamic, and collaborative framework for MM-RSOD, aiming to spark further interest within the community. However, our proposed methods also have some limitations. The modality-specific uncertainty would be more complicated, and it is difficult for our method to tackle noisy information in the frequency domain. Therefore, we will consider how to effectively leverage frequency-aware multimodal information to provide higher-quality MM-RSOD in the future. Moreover, we will consider collecting more complex datasets with scale and imaging angle inconsistencies and conduct experiments on these datasets to verify whether our method can effectively handle such inconsistencies. We will also focus on the trend of changes between the detection performance of our method and the settings of the scale and imaging angle inconsistencies in the dataset, to explore the applicability of the method.

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